

Application of Data Mining to Customer Profile Analysis in the Power Electric Industry

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Abstract—In Japan, the Revised Electric Utility Industry Law went to effect on March 21, 2000. The partial deregulation into the retail electric power supply sector means that electric power for extra-high voltage customers can now be supplied from companies other than the 10 major electric power companies (EPCOs) and electricity rates charged to customers can be determined freely according to negotiations for liberalized sector between these companies and their customers. Due to the introduction of this partial deregulation, the 10 EPCOs have found it necessary to form customer strategies against new entrants, Power Producers and Suppliers (PPSs). In other industries (such as retail trade) in which competition with smaller companies is fierce, it is indispensable to assess preferred customers yielding more profits and form marketing strategies (i.e., business intelligence) to strengthen relationships with these preferred customers by providing unique services. Even in the electric power industry, it is expected to become necessary to form the same marketing strategies as these unrelated industries in consideration of future fierce competition. In order to draw up marketing strategies, it is first indispensable to understand customers, that is, the analysis of customer data. Since the nature of electric power differs from the products that the retail trade business targets, it is necessary to carry out a type of customer data analysis that differs from the retail trade business. The authors explain an example of marketing method to establish customer strategies, using data mining technique based on customer profile data.

Keywords—Business Intelligence, Customer Relationship Management, Customer Strategy, Data Mining, Deregulation.

I. INTRODUCTION

IN the progressing liberalization of the electric power industry worldwide, Japan too revised its Electric Utility Industry Law in May 1999 and began partial liberalization in March 2000. The points of the liberalization were partial retail liberalization targeting extra-high-voltage customers whose power supply is 20kV or above and whose contracted demand is 2000kW or above, and to make it possible for these extra-high-voltage customers accounting for about 30% of the total power demand to freely select their electric power suppliers. In addition, regulations determining electricity rates charged to customers for liberalized sector were removed, so as to enable the free setting of rates according to negotiations between these companies and their customers, and regulations concerning charges for non-liberalized sector were reexamined. The partially deregulated system will be reviewed in 2003 to determine its effects on public welfare issues.

Concerning the liberalized sector, the 10 EPCOs are accessing customers yielding profits by a customer pro-

file analysis in today's competitive environment, and have found it necessary to take measures to prevent customers from switching to new entrants. In this paper, we explain an example of an application of data mining to customer segmentation, as a customer profile analysis for the establishment of business strategies, what is called business intelligence.

II. CONCEPT OF CUSTOMER RELATIONSHIP MANAGEMENT IN POWER INDUSTRY

A. The State of Deregulation in Japan

In August 2001, there were nine PPSs that applied to the Ministry of Economy, Trade and Industry (former Ministry of International Trade and Industry) as new companies in the electric power industry. This was about one year after retail liberalization targeting extra-high-voltage customers started. At present, there are about 50 examples showing that customers changed to new companies entering the electric power business from existing electric power utilities, but this number is very few if we compare the number with the number of customers of existing electric power utilities in terms of contracted electric power quantity. However, there are also examples showing a drop of about 20% in electric power charges by customers changing to new companies, and it should not be forgotten that more customers will change to other electric power sources in the future. In addition, there are many views that targets of liberalization will increase from the reconsideration of the partially liberalized system in 2003. If we suppose the range of liberalization will increase to include high-voltage customers, we can predict fierce competition between existing utilities and new entrants or other major utilities, since customers in liberalized sector should increase significantly compared with the current number.

B. Necessity of Customer Relationship Management in Power Industry

In a liberalized market, Customer Relationship Management (CRM) is very important. CRM involves assessing customers yielding profits and constructing relationships with these customers through the implementation of ideal measures directed at the customers. In other words, although there are market-driven approaches looking at what market segments the traditional marketing serves, CRM is a customer-based approach that continuously offers products aiming at improving customer satisfaction (CS) for each and every customer as well as the idea of one-to-one marketing in the marketing field.

The electric power industry is an industry that had al-

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lowed regional monopolies to date in the background of scale merit. It was unnecessary for electric power utilities to deeply understand customers, since the Electric Utility Industry Law imposed on these utilities the legal obligation to supply electric power to customers. However, in order to effectively oppose new entrants in liberalized sector, it is necessary for these utilities to recognize preferred customers based on customer profiles (e.g., amount of electric power energy consumption, load factor) and not merely view all customers uniformly.

C. Analytical CRM process

In the entire CRM, Analytical CRM process is most important because it supports a full understanding of customers and business strategy planning. Analytical CRM process is consisted of:

- Collect data from any database.
- Reveal hidden information from collected data.
- Gain a complete process for identifying, targeting of most profitable customers.

An example of an application of Analytical CRM process for the electric power industry will be the followings:

(1) Customer Segmentation

CRM, as an electric power company, analyzes customer contract information and company information (type of industry, business scale, capital, etc.), amount of electric power energy consumption, load factor, and customer attributes (from information on customers, e.g., contract, charge payment conditions), and then understands the needs of customers while accessing preferred customers. Partitioning will be performed into finer segments in response to the profits and needs for the electric power companies, where each segment (extra-high-voltage, high-voltage, or low-voltage) is determined by the Electric Utility Industry Law.

(2) Service Menu Planning

CRM offers measures to provide service menus for locking in preferred customers, such as offering discount charge menus and added-value services to what the company considers preferred customers.

(3) Income Analysis

According to the offered service menus, CRM analyzes how both electric power income and related income change, and investigates the contents of the services.

(4) Service Response Analysis

CRM measures whether customers offered services are satisfied with those services, and reflects the results in the service menu planning.

III. CUSTOMER SEGMENTATION USING DATA MINING

Next, we show the idea of customer segments in the electric power industry and the idea of applying data mining to business strategy planning involving segments.

A. Example of Customer Segmentation

First, it is important to judge the value of customers based on the amount of electric power use. What is often given as an example of a measurement method of the value

of customers, is the RFM analysis approach, which carries out analysis focusing on Recency, Frequency, and Monetary value from customer purchasing data in the retail business. However, electricity (i.e., a product in the electric power business) differs from products in the retail industry, and is of a nature of being provided in response to the load demand of a customer during a contracted period. Accordingly, parameters equivalent to the above-mentioned RFM do not exist. Because of this, the RFM analysis approach from the retail industry cannot be applied as is, as a method of analyzing customer data.

Here, we focus on the relationship between the amount of electric power use and the load factor. Figure 1 shows a customer value map based on the amount of electric power use. As shown in this figure, the amount of electric power use and the load factor are each divided into two levels, and all customers are divided into four segments. If we view things from the power supplier, the greater the amount of electric power use, the greater the electric charge income, and the greater the load factor, the smaller the facility investment and operation cost during peak hours. Accordingly, we know that customers for which both the amount of electric power use and load factor are large are preferred customers, and all other customers are regular customers.

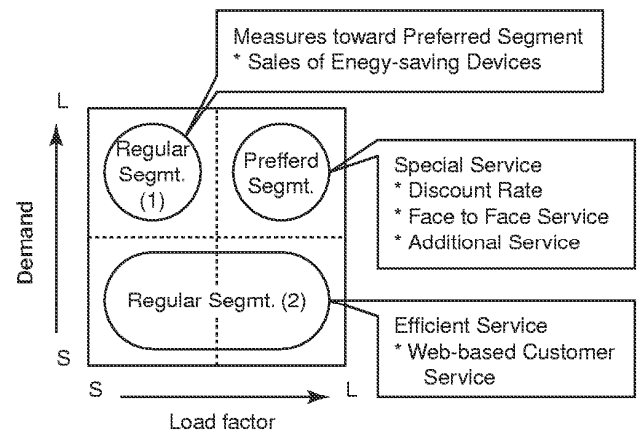


Fig. 1. Example of Customer Segmentation.

We try to lock in preferred customers by offering special services such as discount charge menus, and unique services such as face-to-face services by salesperson. For regular customers, in contrast, we attempt to make our business more efficient by giving priority to answering their needs through call centers and the Internet.

B. Application of Decision Tree

It is necessary to determine what measures to take in response to the preferred and regular customer segments extracted according to the above customer segmentation. For this purpose, we must understand what kinds of customers belong to each segment. As a method of extracting this information from each segment, data mining techniques can be applied. In this paper the authors use a decision tree,

one of the methods most often used in classification and estimation, as our data mining method. The decision tree approach is to construct trees by dividing a population of data and generating nodes through optional explanatory variables selected from individual data attributes, and then classifying the data. Leaf nodes give a classification that applies to all instances and the rules to the class assigned to the leaf. To classify an unknown instance, it is routed through the tree according to the values of the attributes in successive nodes, and when the instance The effectiveness of the classification depends on what standards are used to divide the data. To date, a number of methods have been proposed. The most basic method is a method that selects explanatory variables so that the entropy defined by the following equation is minimized by the division.

$$Entropy(S) = \sum_{i=1}^N -p_i \log_2 p_i \quad (1)$$

Here, S is the whole population, and p_i is the ratio at which class i corresponds to S .

A decision tree can create rules to explain the variable of the judgment target via the other explanatory variables, for one of the variables in the population that each data holds.

C. Example of Customer Strategy

We show the idea of applying the decision tree to the establishment of business strategies for each customer segment. Here, we use the decision tree, and we exclude the load factor from the parameters (concerning preferred customers in which the amount of electric power use and load factor are extracted as parameters) as explained above. Then, through the decision tree, we learn the rules for extracting the segments belonging to preferred customers from among all customers. According to the generated rules, we then understand with what parameters (other than the load factor) preferred customers can be characterized. For example, if we can learn that the possibility is high that there are more than 3,000 preferred customers in the manufacturing industry, we can consider services to create discount charge menus specialized to a large part of the manufacturing industry. The same rule extraction approach can also be effective for regular customers. For example, if we can learn that there are a number of customers that do not have specified energy-saving devices such as storage devices, we can consider a policy to carry out business activities so as to carry out the sales promotion of these devices.

Next, we explain the time changes of customer segments. We have to check the response of customer service offered to each segment. In addition, we compare two time cross-sections, implement customer segmentation on each of the time cross-sections, shown in Figure 2, and then focus on the changes of the customer segments. The following policies can be considered depending on whether or not there are movements between customer segments.

- When there are no movements between segments
→Continue the implementation and use it for the preferred segments.
- When there are movements between segments
→Investigate whether or not there are changes in the load facilities, e.g., the introduction of new facilities.

A decision tree method is also effective to investigate the attributes of each segment and the changes between segments.

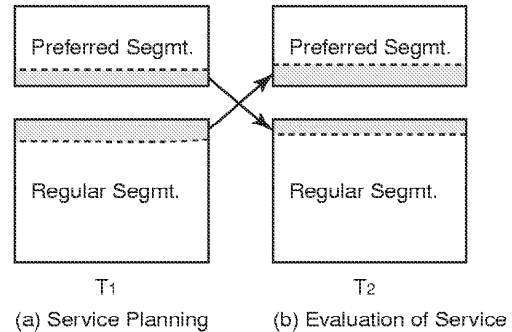


Fig. 2. Transition between Customer Segments.

IV. CONCLUSIONS

In this paper, under the deregulated electric retail market, we stated that it has become necessary for major electric power companies to have retail strategies including service menus based on customer profile analysis. We also described a customer segmentation method becoming the foundation of one-to-one marketing, and a business strategy establishment method using data mining. In the future, we plan to proceed with investigating more detailed customer profile analysis methods.

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VI. BIOGRAPHIES

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